Experiment Name: Basic Building Blocks in Python programming language

In lab report:
In code section solve this problems for code.

General Instructions for writing any lab report:

- Use Times New Roman font.
- Font size: 12 pt.
- Line spacing: 1.15.
- Use justified alignment for the main text.
- Use appropriate headings and subheadings.
- Write in clear, formal, and academic language.
- Include only the required code for solving each problem. Do not include algorithm or flowchart unless specifically requested.

The lab report must contain the following sections in order:

1. Experiment No.
    - Mention the experiment number.
2. Experiment Name
    - Provide a clear and descriptive title of the experiment.
3. Introduction/Theory
    - Explain the theoretical background relevant to the experiment.
    - Introduce the data mining concept, technique, algorithm, or software used.
    - Describe why the experiment is important and where it is applied.
    - Keep the explanation concise but informative.
4. Objective
    - State the learning objectives of the experiment.
    - List the skills or concepts that students are expected to learn.
5. Problems

For each problem, create a separate subsection using the following format:

Problem 1:

- Problem Statement (if provided)
- Code
    - Write a complete, properly formatted, and well-commented program that solves the problem.
- Input
    - Show one or more sample inputs.
- Output
    - Show the corresponding outputs produced by the program.

Repeat the same structure for every remaining problem:

Problem 2:

- Code
- Input
- Output

Problem 3:

- Code
- Input
- Output

Continue similarly until all problems are completed.

6. Discussion
    - Discuss what was learned from the experiment.
    - Explain how the implemented programs or techniques work.
    - Mention any observations, challenges faced, limitations, or practical applications.
7. Conclusion
    - Summarize the overall outcome of the experiment.
    - State whether the objectives were achieved.
    - Briefly mention the knowledge and practical skills gained from completing the experiment.

Additional Guidelines:

- Ensure every code section contains a complete working solution.
- Maintain proper code indentation and readability.
- Use meaningful variable names.
- Include sample input and output for every problem.
- Keep explanations concise and academically appropriate.
- Ensure consistency in formatting throughout the report.
- Do not omit any of the required sections.

Problem 1: Temperature and Currency Converter

Problem Statement:

Write a Python program that accepts a temperature in Celsius and an amount in USD. Convert the temperature to Fahrenheit and the currency to Bangladeshi Taka (assume rate: 1 USD = 110 BDT). Display both converted values.

Sample Input:

37.5

25

Sample Output:

Temperature = 99.50 F

Amount = 2750.00 BDT

Problem 2: String Search and Censorship

Problem Statement:

Write a Python program that takes a sentence, a target word to mask, and a search keyword. Replace all occurrences of the target word with *** using string replacement, and display whether the search keyword exists in the original sentence along with its index position.

Sample Input:

The quick brown fox jumps over the lazy dog

fox

quick

Sample Output:

Masked Sentence: The quick brown *** jumps over the lazy dog

Keyword 'quick' found at index: 4

Problem 3: Multiples Generator and Summation

Problem Statement:

Write a Python program that reads an integer N and a multiplier K. Print all multiples of K from 1 to N, along with the total count of multiples and their sum using a for loop.

Sample Input:

25

5

Sample Output:

Multiples: 5 10 15 20 25

Total Multiples = 5

Sum = 75

Problem 4: Word Frequency Counter

Problem Statement:

Write a Python program that accepts N words, stores them in a list, and converts them into a dictionary where the keys are unique words and the values are their occurrences. Display the frequency dictionary and the word with the highest count.

Sample Input:

6

apple

banana

apple

cherry

apple

banana

Sample Output:

Frequency Count: {'apple': 3, 'banana': 2, 'cherry': 1}

Most Frequent Word: apple (3 times)

Problem 5: Student Gradebook and Above-Average Filter

Problem Statement:

Write a Python program that takes input for N students with their respective marks and stores them in a dictionary. Calculate the class average score, and display all students who scored strictly above the average mark.

Sample Input:

4

Anik 78

Bipasha 92

Chayan 64

Dipa 88

Sample Output:

Class Average = 80.50

Students Above Average:

- Bipasha: 92.0
- Dipa: 88.0